

Target Information and Profile System (TIPS) Recommendation and Evaluation of Future Direction

Final Report

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Prepared by

Yat-Sang Hung
Member of Technical Staff
U S WEST Advanced Technologies

Abstract

This document provides detailed descriptions and analysis of user problems, objectives, and requirements for the Target Information and Profile System.

It also provides recommendations for the future direction of that system and final feedback from the readers.

Version Notice

All revisions made to this document are listed below in chronological order. The release number of this document does not relate to the release number of the application:

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Yat-Sang Hung
U S WEST Advanced Technologies
1475 Lawrence Street, 4th Floor
Denver, Colorado 80202
(303) 595-0826

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The quality and utility of the knowledge base required to implement the sales helper is evaluated according to the following criteria:

- nature, form and extent of knowledge about the relationships between product recommendations and customer information and needs;
- source and validity of such knowledge.

No quantitative relationship has been found. Our investigation found that the only knowledge is the qualitative and intuitive judgement of the subject matter experts (CCEs and marketing staff). Given the lack of easily elicitable and robust knowledge, any engine, regardless of technology, cannot be expected to perform very well in the initial phase.

1.3 Recommendation for TIPS

Functions

Phase 1

- collection of pertinent customer information for marketing and CCEs
- customized recommendations generated from customer needs and profile
- a value-added sales assistant module including a subset of the following
 - automation of SIC entry (in English), expansion of SIC to include short descriptions of business procedures, advertisement, blue-light special, office blitz, new product announcement, directory closing date, local calling areas, tax tables for each area, central office and equal access conversion dates, personal sales and revenue profile of CCEs

Later Phases

- customer telecommunication usage and traffic patterns in later phases of the project (core functionality to support product recommendations)
- toll usage or bill amount for last 12 months
- improved product recommendations when the quality of the knowledge base for customer information and market conditions improves; account usage and traffic pattern of customers will be used as one of the major bases or determinants of generating recommendations

Technology

- expert system architecture flexible enough to allow future expansion and hooking of statistical or neural network modules

Strategies

- involvement of 4 - 6 subject matter experts from CCEs, with subject matter experts from marketing (both owner/operator and manager market) for two 1-day workshops to set up scope and requirements of knowledge acquisition process (first workshop) and cross validation of knowledge (second workshop)
- knowledge acquisition from subject matter experts (2 from CCEs and several from marketing, representing owner/operator and manager market and product management); knowledge validation and enhancement sessions of the subject matter experts
- resources from marketing allocated to study telecommunication needs and product usage of different industries (for example, along SIC dimension) if recommendations are to be generated at that level of granularity as envisioned by the management
- resolution of issues between organization/management-originated and user-originated requirements
- statistical analysis tool for market research and analysis in SBS

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2. Introduction

2.1 Purpose

This article documents the user requirement collection process of Target Information and Profile System (TIPS) and some related issues which will impact the future direction and usability of the system.

The overall objective of TIPS is to organize, represent, and present sales, product and marketing knowledge in a form that can be shared, collected, communicated and utilized effectively by Customer Contact Employees (CCEs) and other departments involved.

In the initial design and implementation, TIPS is envisioned to be an intelligent sales tool integrated seamlessly with Sales and Service Negotiation and Quality (SSNQ) system. It will generate customized product recommendations, benefits, and counters based on the characteristics of the customers and other information deemed relevant.

Since the problem domain of TIPS is complex and difficult to define, it is important to decide the feasibility and necessary conditions for TIPS to succeed. This report is an attempt to propose a set of semi-structured criteria to establish and evaluate the possibility of the knowledge base to implement the intelligent sales tool. It also proposes some preliminary procedures and guidelines to evaluate the usability of the knowledge base derived from the subject matter experts.

The report will not address the details of information needs of CCEs collected in the user requirement process. However, it will highlight some of the important requirements and put them into perspectives.

This report also identifies some of the issues between the end-users (CCEs) and the management. These issues may need to be addressed before a successful system can be developed. A set of recommendations is also made to the technology options and future direction of TIPS.

2.2 Author

The author of this document is Yat-Sang Hung, Member of Technical Staff, Standard Services and Human Factors Development, US WEST Advanced Technologies. Documentation support was provided by James Beeter, Information Development, Information Technologies (IT).

2.3 Disclaimer

The contents of this document represent only the views of the author. This document is intended to be used as a reference to the future direction and usability of TIPS or similar systems. The findings of this report are not meant to be used or interpreted outside the scope of TIPS or similar systems.

2.4 Acknowledgments

The author is grateful for the help of the following individuals for their valuable advice and input: Ed Barney, Judy Warner, Cathe Valdez, Gary Craig, Susan Wagner, Gail Fontenelle, David Sward, Jim Beeten, Chris Plott, Mike Henry, Alan Lawson and Kathy Eckman.

2.5 Project Background

The concept of having an intelligent sales tool can be dated back to the Home and Personal Services (H&PS) Systems Vision Requirements in 1991. In that document, it was mentioned that artificial intelligence could be used to recommend products/services based on household profile. That document also contained snapshots of customer profile screens as well as product/service recommendation screens. Two proposals, *Marketing Vision* and *Integrated Vision* were released (1,2). Prototypes of *Marketing Vision* and *Integrated Vision* were also developed.

The intelligent sales tool was eventually incorporated into the prototype of CLINK Sales Assistant (CLSA). CLSA was designed and implemented by Carnegie Group Inc. When CLINK was cancelled, the intelligent sales tool was revamped and implemented as the Sales Assistant (SA) for demonstrating the concept of intelligent sales tool. The concept of generating recommendations from customer profile was also incorporated in the prototype of Service Activation Control Systems for H&PS and SBS produced by US WEST IT and AT (3). When SACS failed to materialize, the concept of the intelligent sales tool was included in the business case and user requirement of Sales and Service Negotiation and Quality (SSNQ) (4).

At about the same time, Carnegie Group used the Sales Assistant and reimplemented as CASH (Consultation and Service Helper) and deployed in Salt Lake City as the Home-Office Trial (HOT) for H&PS and SBS. The project was a simple test to see if the Home Office Market was accessible and had the attributes that US WEST market analysis found. It was done with a shortage of time and resources. According to Alan Lawson, Director of Engineering of the Knowledge Based Technology Development of CGI, the project fulfilled its purpose of testing the Home-Office Market, though there was not enough resources and time to address the human

factors side of the system. Field evaluation of the program by human factors engineers at the early stage of the trial (November, 1992) revealed some limitations of CASH (5). Among the most important is the ineffectiveness of the rules to provide sales assistance for the experienced consultants. The system was not integrated with the negotiation system because it did not take existing services into account (no data services were implemented because of lack of budget and time). The list for customer profile was so long that the users did not have time to complete it. After the release of the usability test, Carnegie Group corrected some of the usability problems and gained some valuable insights of the sales tool from the trial. According to the announcement of Don Baltzer (H & PS) on May 19, 1993, the creation of the Home Office Market out of H & PS and SBS and its approach to the market grew out of HOT.

The consultative sales help requirement was incorporated into the requirements of Sales and Service Negotiation and Quality System (SSNQ) for Small Business Services (SBS) Customer Contact Employees (CCEs) to provide quality consultation services to customers. The module was renamed as Customer Analysis and Sales Helper. It was renamed again to Target Information and Profile System (TIPS) to avoid confusion between the system and CASH in the Home Office Trial.

Because of various reasons such as the availability of resources and time, the initial design and the architecture of SSNQ did not take into consideration the requirements of TIPS. We will work with the SSNQ team to determine the schedule of TIPS in the context of SSNQ.

Carnegie Group Inc. has been involved in the sales tool since its inception. If expert system technology and Carnegie Group Inc. is chosen as the vendor for TIPS, Kathy Eckman with her development team from CGI will design and implement the underlying engine and knowledge base of the system.

2.6 Problem Domain, Project Scope and Constraints

Although the concept of an intelligent sales tool has been around for quite some time, the problem domain is relatively vague and ill-defined. One of the major challenges of this project is to develop an intelligent sales tool that can generate quality service and product recommendations based upon customer profile information and needs.

While it is easy to list the ideal functional requirements of such a sales tool, it is not at all clear that the current marketing data, information or knowledge can support the construction of such a tool. In order to generate a non-trivial recommendation, knowledge (implicit or explicit, declarative or procedural) of the relationship between customer characteristics and product recommendations has to be built into the knowledge base of the

engine of the tool, no matter what technology is used to implement the engine. Given the fuzziness and difficulties in this area, the following questions should be addressed to study the feasibility of such a project:

1. What is the nature, basis and form of the knowledge? At what level of "granularity" can the knowledge be organized?
2. If such knowledge is possible, does the knowledge (not data) currently exist? If not, how much effort and time will be expended to acquire such knowledge?
3. Is the knowledge exact or fuzzy? Can we assign some kind of degree of confidence in the accuracy of such knowledge?
4. How can we validate the knowledge?
5. What and how will the knowledge change over time?
6. What pertinent information is needed to build up or enhance such knowledge?
7. Where/who can we extract the knowledge from?
8. How can we extract and/or make use of the knowledge?
9. What technology can be used to develop a system utilizing the knowledge captured? How?
10. Assuming that the knowledge is obtainable and usable, is it common-sense or esoteric for the users?
11. Assuming that the knowledge is obtainable and usable, how can the users apply such knowledge easily and effectively?

The matrix below shows a proposed mix of responsible parties for the answers of those questions (the columns show the question number):

	1	2	3	4	5	6	7	8	9	10	11
SBS	PRI	PRI	PRI	PRI	PRI	PRI	PRI	SEC		SEC	SEC
CGI	SEC			SEC		SEC	SEC	PRI	PRI	SEC	SEC
IT									SEC		SEC
AT	SEC			SEC		SEC	SEC	SEC	SEC	PRI	PRI

PRI: Primary Responsible Party

SEC: Secondary Responsible Party/Parties

Most of the above questions can be expanded upon. For the purpose of this document and the user requirement document, we will focus on the first 7 questions in the requirement collection and analysis process.

Before we address the details of the user requirement collection process and its results, we would like to define the scope of our endeavour.

The scope of the project includes the following:

- conduct knowledge acquisition sessions with the subject matter experts to elicit the knowledge necessary to build the knowledge base for the engine; it is assumed that the subject matter experts will be able to supply the necessary knowledge in a timely and consistent fashion;
- make suggestions or recommendations to the type and design of research to build knowledge necessary for the knowledge base; the decision and actual execution of the research is the responsibility of the client organization;
- organize the knowledge elicited from the subject matter experts and other sources (e.g. product statistics) and implement a system capable of generating recommendations from such knowledge using an appropriate technology such as expert system technology;
- provide a simple interface where knowledge can be updated easily by the client organization;
- incorporate some components (to be agreed upon) of the recommended sales assistant module to add value to the system;
- design user interface prototype for all the components of TIPS; implementation of user interface will be provided through SSNQ;
- specify the type of data services needed; implementation of data services (to legacy systems and CSDBs) will be provided through SSNQ;
- conduct usability test and evaluation on the knowledge base and the user interface of TIPS.

Because no quantitative relationship can be found, the knowledge base will be constructed from the qualitative and intuitive judgement of the subject matter experts. Quantitative knowledge derived from the analysis of data collected in the subsequent phases of the project may be incorporated into the knowledge base eventually.

The scope of the project does not include the following:

- plan or conduct marketing research for SBS to determine the relationship between product recommendation and customer information and needs as defined above;
- plan or conduct marketing or any related research to build or enhance knowledge for the knowledge base;
- maintain any changes in product information, advertisements, office blitz, blue-light specials and any other data that need updates after system is implemented;
- conduct any research on the data that TIPS gathers.
- interface with any legacy systems and CSDBs.

2.7 Assumptions and Dependencies

It is assumed that SSNQ can provide the necessary resources for data services and user interface component development (programming) in a timely and consistent manner between July and November, 1993. Lack of such support will have an impact on the quality and schedule of the project. The amount of resources needed will be estimated formally after the engineering team of Carnegie Group Inc. finishes their first review of the user requirements.

Besides SSNQ, the success of TIPS also depends on the quality of the knowledge base used. The quality of the knowledge base, in turn, will depend on the following:

- the knowledge of subject matter experts;
- the knowledge acquisition process and technique;
- the applicability of the underlying knowledge representation scheme and the engine.
- time and resource constraints of the project.

Although there is no foolproof method to ensure those conditions are optimal or achievable, suggestions will be made to the strategies of enhancing and maintaining the quality of the knowledge base in the process. The first and the last conditions shown above are beyond the control of the project team of TIPS.

2.8 Knowledge Base Usability Evaluation

In order to improve the quality of the knowledge base and the rules to generate recommendations, the following plan can be implemented to ensure continuous improvement and evaluation:

1. Subject matter experts (CCEs, marketing and product development) together with the knowledge engineers of Carnegie Group (if CGI is selected) will define the products that need to be included in TIPS (early or mid June). The salient information about the customers obtained from knowledge acquisition process will be mapped back to these products.
2. Extraction of knowledge from subject matter experts takes place between June and October; a review meeting with subject matter experts representatives will be held each month to validate the knowledge collected. Tentative schedules for knowledge acquisition sessions will be mid-June, early July, end of July, mid-August and early September. Each session will take about 2 days or 3 days mostly in Phoenix and may involve different subject matter experts across SBS.
3. After the contents of account/customer profile/needs are determined by the subject matter experts, a set of scenarios of sales consultation will be developed from independent subject matter experts not participating in the project and the subject matter experts from which the knowledge is extracted from. Product recommendations will be generated from them to serve as testing material for the engine and the knowledge base. This can be developed between July and October.
4. The prototype engine will be tested against a subset of the scenarios prepared from both groups to find out the accuracy and the effectiveness of the recommendations. The purpose of having two groups is to enhance the robustness of the knowledge base. Disagreement of recommendations between 2 groups can be addressed and problems can be uncovered.
5. If time permits, iterations can be conducted to refine the performance of the prototype and tests can be conducted with different subsets of the scenarios.

If time and circumstances do not permit, the above procedure may be simplified to the construction of scenarios by the participating subject matter experts. There may be some trade-off in the validity of the results of the tests since the same subjects are supplying the "rules" of the knowledge base. However, using scenarios to validate the knowledge base is important as a check-point of the knowledge obtained. Initial testing can be done in September or October 1993.

Assuming that the data services from SSNQ are available before October, 1993 for TIPS, knowledge base usability tests can be conducted in the living laboratory environment in Denver. CCEs participating in the tests can be asked to rate the effectiveness and usefulness of the sales tips generated from TIPS. Differences between the tips generated and the actual products that the customers buy can be compared. Differences between the tips generated and the CCEs own set of recommendations can also be collected for analysis.

Given the complexities of the domain, it is advisable to keep the test scenarios as real-life as possible. Thus, the scenarios can be constructed from real customer data, e.g., 100 customers can be selected at random and interviewed by CCEs to determine the parameters used to generate recommendations and the recommendations that the CCEs think most appropriate. The information of the products the customers intend to buy after hearing the recommendations can also be collected. Again, if resources do not permit, "fictitious" scenarios can be developed for the purpose with some trade-off in the accuracy of the test results.

Another time and resource saving alternative will be using a panel of subject matter experts to evaluate the recommendations generated from specific customer profile and needs information. This is much less rigorous than the approach adopted above as it does not involve the construction of the entire scenarios beforehand and the verification of what the customers intend to buy after hearing the recommendations.

It is important to note that such evaluation is conducted to improve the quality of the knowledge base and to facilitate participation of users in the knowledge acquisition process. They can neither directly increase dramatically the amount of "knowledge" of the subject matter experts nor accelerate the pace of the knowledge acquisition process.

The criteria for acceptance of the knowledge base has not been set. The metrics needed to measure the validity, reliability and accuracy of the knowledge base will have to be developed and agreed upon. As a first cut, we can approach the evaluation at two levels - the individual phone line level and the overall product recommendations level. Because of the nature of the problem domain, we will not expect a very high hit rate on the recommendation generated. Individual differences of the subject matter experts may also contribute to differences in recommendations. In the initial phase of the project, a hit rate of over 70% or within 10% of the consensus of the subject matter experts will be reasonable. As the quality of knowledge base improves, a higher hit rate can be expected.

3. Methodology

An eclectic approach is used to gather user requirement information. Because of the difficulties and time in scheduling extended meetings with the users and management team of our clients, we have to be very flexible in our tactics and form of the meetings. As an example, the focus group destinations, sizes and times have been changed several times.²

Another limitation to the user requirement information gathering process is the constraints prescribed by upper management. Meetings with the CCEs were restricted to feedback on the presentation of TIPS. CCEs were not allowed to have any input to the contents or functions of the proposed software developed for their use. The participants of the focus groups were selected by management according to performance level. Three groups were used (high and medium performers and new CCEs for two cities and all high performers were used in one city). Because of the nature of the selection process and the mix of Tier-1 and Tier-2 cities, it would not be possible to generalize any findings. Caution has to be exercised in interpreting the results.

There are several broad categories of information-gathering methods - questionnaire and survey, interviews, observation, workshop, and focus groups, as well as research into the secondary source materials like past proposals, business plans etc. Questionnaires and surveys were given a low priority because of the expected poor response rate. People in general, and CCEs in particular would tend not to complete surveys and questionnaires we send out. In order to maximize our use of time with the CCEs, the focus group format was proposed by Ed Barney. With the help of a professional facilitator, the focus groups proved to be a worthwhile methodology, even though they were conducted in circumstances that were far from ideal.

² The focus groups of the service representatives were originally scheduled to take place in Minneapolis (Mar. 4 and 5), Seattle (Mar. 15 and 16), Portland (Mar. 17 and 18) and Phoenix (Apr 1 and 2) with 6 persons in a group and 2 to 3 groups in a city. Because of access and various other reasons (e.g. training, labor...), the location and time changed to Portland, Omaha, Denver and Salt Lake City in early April. However, the service centers experienced access problems and the number of participants of the focus groups reduced to 2 or 3. Salt Lake City and Denver focus groups were cancelled and the time was changed for the visits to the other 3 cities. We completed our focus groups in Portland, Omaha and Albuquerque (total of 21 CCEs).

3.1 Interviews

Interviews are one of the most informative methods to elicit user requirements, particularly when those interviewed receive a copy of the questions to be answered beforehand, so that they can make adequate preparations. We employed this strategy when we conducted our interviews with Greg Winn (Director of Sales of SBS, executive sponsor of TIPS), Frank Pereira (Director of Business Development, SBS), Ron Rinkle (Director of Owner/Operator Market, SBS), Bill Stack (Manager of Owner/Operator Market, SBS), Steve Archuleta (Director of Systems, SBS), Phyllis Lopez (Manager of Manager Market, SBS). Enclosed are the questions used for the interview. These interviews took place between Feb 10 and Feb 13, 1993 in Phoenix. The results of these interviews formed the major portion of organization/management-originated requirements.

Another set of interviews was conducted with the marketing staff. Again, a set of interview questions was distributed before the interviews to set the focus during the sessions. The interviews took place in Phoenix between May 5 and May 7. Participants included Laird Brown (Owner/Operator Market, SBS), April Neiding (Owner/Operator Market, SBS), Bryan Meincke (Manager Market, SBS), Mitchell Glasser (Market Research and Analysis, Strategic Marketing, US WEST Inc.) and Beth Treadway (Product Analyst, Strategic Marketing, US WEST Inc.).

3.2 Focus Groups

The use of focus groups is another informative method to elicit user requirements. The relatively relaxed atmosphere with the emergent positive group dynamics facilitates the users to express their ideas and requirements openly. Though the circumstances under which the focus groups were conducted were far from ideal, we still established open communication and collected robust user requirements. The number of participants per group was reduced from 6 to 2 or 3. We (5 persons including the facilitator) also were in the same room. Originally, we thought of having a separate room and observing the participants through a one-way mirror or videotaping the sessions. However, we also felt that it might breed mistrust and create more artifacts in the focus groups. We decided to join the focus groups instead. Because of the confidentiality agreement, the participants seemed to be very open and candid, even in the presence of the team members of TIPS (Cathie Valdez, Kathy Eckman and myself) and a representative from SBS management (Dale Crum, Assistant Manager, Sales Operation).

Focus groups were held in three cities, Portland (Tier 1, Western), Omaha (Tier 2, Eastern) and Albuquerque (Tier 2, Central) between April and May respectively. Three focus groups (high performers, medium performers and new CCEs) were conducted (for each city) with 3 participants in each group for Portland (Tier 1) and 2 participants for Albuquerque (Tier 2). Omaha has the same number of participants in the focus groups but all three groups are high performers. All these groups were mandated to be selected by management.

3.3 Subject Matter Expert

The project manager of TIPS, Cathe Valdez, also acts as the subject matter expert for the CCEs. She has more than 15 years of experience in the business office. Mike Henry, the client lead for SSNQ, will also act as another subject matter expert for the CCEs. Laird Brown is the tentative point of contact and subject matter expert on marketing, though it is highly possible that his role may change in the near future. More subject matter experts will be expected to get involved in this project to build a more robust knowledge base for the system. We will work with them closely to elicit their knowledge on the area necessary for the success of the project.

3.4 Other Research

Other sources of research include past functional requirement documents, business plans, past prototypes of related projects and systems. The most recent research done as a prelude to TIPS are the CASH Evaluation Trial, Salt Lake City, Utah and Consultative Behavior: Knowledge Engineering Report and Analysis prepared by David Sward and Gail Fontenelle of US WEST Advanced Technologies (6). Most of the findings in this report corroborate with their results.

Preliminary analysis of the suitability of statistical data as a predictor of product usage pattern from functional group or SIC by marketing showed that there is no relationship (7). Besides, most CCEs rank those data as medium in usefulness and marketing staff also remarked that they cannot find any consistent relationship of product purchase pattern with functional groups.

Table 1. Summary of the TIPS Framework	
Component	Description
1. Vision	Define the long-term goals and objectives of the organization.
2. Strategy	Develop a plan to achieve the vision, including identifying key areas of focus and allocating resources.
3. Structure	Design an organizational structure that supports the strategy, including defining roles and responsibilities.
4. Culture	Foster a positive organizational culture that aligns with the vision and strategy.
5. Leadership	Establish a strong leadership team that guides the organization and inspires employees.
6. Talent	Attract, develop, and retain top talent to drive organizational success.
7. Technology	Invest in technology to enhance productivity and innovation.
8. Financials	Manage the organization's financial resources effectively to ensure long-term sustainability.
9. Risk Management	Identify and mitigate potential risks that could impact the organization's performance.
10. Performance Measurement	Establish key performance indicators (KPIs) to track progress and evaluate success.
11. Communication	Ensure clear and consistent communication across all levels of the organization.
12. Innovation	Promote a culture of innovation and encourage employees to think creatively.
13. Sustainability	Integrate sustainable practices into the organization's operations and strategy.
14. Compliance	Ensure the organization adheres to all relevant laws, regulations, and industry standards.
15. Social Responsibility	Engage in social responsibility initiatives that benefit the community and enhance the organization's reputation.

4. Results and Discussion

4.1 Critical Success Elements of Consultative Selling

The primary critical success element of consultative selling is establishing rapport with the customers. Its importance was reflected in all the focus group sessions with CCEs. Virtually all participants stipulated that it is a necessary condition of successful customer contacts. In general, rapport can be facilitated by the following strategies employed by CCEs:

- engage in active listening to the situations of the customers - it serves as a basis to learn and understand customers' needs; besides, it shows that CCEs are willing to spend time and effort to serve customers
- show understanding and genuine concern for the customers and their business needs - it creates trust and paves the way to long term relationships and future growth
- solve telecommunication problems of the customers effectively and efficiently - it builds customers' confidence in the competency of CCEs and establishes superior reputation
- provide appropriate solutions and recommendations to the customers that will satisfy their needs - it enhances customers' perception of CCEs' professionalism; instead of providing predetermined cut-and-dried solutions, CCEs respond to individual needs and customize the recommendations
- provide education and explanation of the products offered and how these products satisfy the business needs of the customers
- offer apology and personalized assistance to dissatisfied customers

These findings are consistent with the *Customer Needs Analysis* of US WEST Communications, Inc. SBS (B). In the survey, being flexible, understanding and responding to customers' individual needs is found to be the most important customer requirement, followed by making commitments that meet customers' needs.

Another important element of success is the availability of relevant and pertinent customer information to support consultative sales activities. In many consultative sales situations, it will be difficult for the CCEs to convince the customers without substantiating the reasons why a particular set of product recommendations is made.

One of the best sources of information (indicated by the CCEs in the focus groups) is the usage, traffic and toll information (including customer current records and current bill) of the phone lines of customers. Traffic and busy studies of customers' lines including past call patterns, toll and bill

information is valuable information that CCEs can utilize in making recommendations to customers. This can be presented as objective and hard evidence (empirical and historical information) to convince customers accepting CCEs' analysis and advice on their telecommunication needs. In conjunction with the analysis of needs from the customer profile information, it will form the core of the strategic information CCEs can use.

4.2 Organization/Management-originated Requirements

The objectives of the system, as delineated by the management are

- to sell deeper and broader into the market, that is, help CCEs to sell across the entire range of products that US WEST offers small businesses and enable them to sell more products in a single contact
- to consult professionally with the customers for value-added solutions from US WEST to meet their business needs
- to maximize revenue, productivity and efficiency of any sales contact
- to support the WRAP strategy

4.3 User-originated Requirements

The end users of the system will be the CCEs of SBS. To develop a successful system that the CCEs will use to fulfill the mission of the company requires extensive user participation in the design of the system. Besides, their field experience is indispensable in the construction of a robust and accurate knowledge base of the market and the customers. They should form one of the primary sources of knowledge acquisition in the building of the system.

One obvious requirement of such a system is the capability of providing support to the CCEs so that they can concentrate on the development of their expertise in consultative selling. It should not stifle internal motivation of the CCEs by taking away the enjoyable part of their work. Highly motivated CCEs enjoy using their expertise to solve telecommunication problems for their customers. They need to know the unique business circumstances of their customers as well as their usage and traffic patterns of their phone lines to generate and substantiate their recommendations.

Other suggestions include:

- advertisement, special and office blitz information
- daily feedback for CCEs
- account profile information
- expansion of SIC to include description of business procedures
- future product announcement
- cue words for product benefits and counters instead of statements of benefits and counters
- automation of SIC code and use of English to enter code

4.4 Organization and User Issues

The following issues are largely management issues. Their effect on the design and implementation of TIPS will be determined by the management. The author does not have any stance or position in these issues.

There are several possible incompatibilities:

- implementation of WRAP strategies - CCEs stipulated that each customer is unique and they make recommendations based on customers' needs, not out of packaged recommendations driven by compensation. Virtually all CCEs in the focus groups mentioned and agreed on that. The quota imposed could have a negative effect on customer service as CCEs push harder to meet their objectives with the customers and possibly create a negative impact on customers.
- use of functional groups, SIC and market segment information - CCEs made little use of the information to make recommendations and deemed that they were not useful in practical situations.
- the past and current office objectives do not support consultative behavior. Often the pressure of meeting objectives outweighs the importance of conducting a thorough consultative contact.

4.5 Problem Domain Analysis

An intelligent sales tool that generates recommendations from customer information and needs can only be as good as the pertinent knowledge on which it is based (declarative or procedural, implicit or explicit), independent of the technology it uses to implement the tool. With this theme in perspective, we will consider the rationale of the questions and the corresponding answers to those questions from SB5:

1. *What is the nature, basis and form of the knowledge? What level of granularity can the knowledge be organized?*

Rationale:

This question attempts to identify the underlying knowledge model of the users. Knowledge of the relationship may be empirical or non-empirical or a mix of both. Empirical knowledge can be derived from the study of past historical data like the distribution of specific products among specific functional groups or occupation/profession. Non-empirical knowledge (or

semi-empirical knowledge) may be derived from "gut-feelings" of subject matter experts (expert opinions based on experience); qualitative analysis of customers' needs... Knowledge may be declarative or procedural or both.

It will be very useful if the knowledge is detailed (specific needs of specific class of customers can be related to specific products).

Response:

According to market research and analysis personnel in SBS, Bryan Meincke (Manager Market), Laird Brown (Owner/Operator Market), they cannot find any quantitative relationships between customer characteristics, market conditions and the telecommunication services bought. Bryan has done numerous studies on the data of Manager market, including distribution of products/ services with 1 and 2-digit SIC Code, product utilization per CLAC and per line. The only pattern he identifies is the higher number of listings bought by legal and health professionals.

Mitchell Glasser (Market Research Manager of Strategic Marketing of US WEST Inc.) remarked that there was no quantitative knowledge of the above mentioned relationship.

The knowledge we can make use of in building the system is qualitative analysis of customers' needs in relation to the products and services. The project DARWIN is a study of the market and products from the customers' perspectives. More about DARWIN will be known in June. The results of DARWIN may have an impact on SSNQ and TIPS in the future.

2. Is the knowledge mentioned above possible, given the time and resource constraints?

Rationale:

This can be used to estimate how much effort and time required if we need to obtain such knowledge.

Remarks:

Quantitative analyses will be time and resource consuming. According to Mitchell Glasser, it is very difficult and cumbersome to get data from the downstream systems. Besides, specific customer profile and business information is not available. Since there is no discernible pattern of the quantitative data for the engine, focus should be shifted to qualitative analysis of customers' needs to acquire the necessary knowledge. The time and effort required is difficult to estimate as it depends very much on the knowledge of the subject matter experts and their readiness to articulate that knowledge.

3. *If such knowledge is possible, does the knowledge (not data) exist as of today?*

Rationale:

If it exists, sources of such knowledge can be identified and incorporated into the development of the system.

Remarks:

Qualitative knowledge exists but is "distributed" among the subject matter experts, market analysts, product managers and CCEs. In the knowledge acquisition process, we need to merge the distributed pieces of knowledge together. This will form the basic challenge of TIPS in the initial phase of the project.

4. *Is the knowledge exact or fuzzy? Can we assign some kind of degree of confidence in the accuracy of such knowledge? Can the knowledge be validated and how can we validate it?*

Rationale:

A system is only as good as the knowledge it has. One criticism leveled against the CASH prototype of H & PS is the accuracy of recommendation it generates. Given the time and circumstances under which it was implemented, it was apparent that it would suffer from those problems. If we depend on only one or two subject matter experts for information, we may miss important alternatives or aspects.

Remarks:

Knowledge of this nature is not exact. To enhance the validity and accuracy of knowledge obtained, we have to involve more subject matter experts and conduct cross validation sessions among marketing personnel, product managers and CCEs. Knowledge base usability evaluation (or its variation) as outlined in this paper can be conducted.

5. *What and how will the knowledge change over time?*

Rationale:

A tool is useless if it cannot be changed or updated as knowledge of market and demographic conditions change. New methods of analysis and research will yield new knowledge. It should be incorporated into the tool continuously to keep the information competitive.

Remarks:

Knowledge will change and accumulate over time; the engine should be able to accommodate different types of knowledge (quantitative as well as qualitative). Besides, new perspectives of analyzing the market may change the knowledge drastically (e.g., DARWIN may bring about some changes in the future).

Eventually, a knowledge editor must be provided as an interface to maintain the knowledge base of the system.

6. *What pertinent information is needed to build up or enhance such knowledge?*

Rationale:

The needed information can be gathered through TIPS. However, given the demanding environment of CCEs, such information should be kept to a minimum.

Remarks:

This will form part of the agenda for the knowledge acquisition process. We may have to perform some analysis to consolidate and streamline the needed information. Such procedures will have to be investigated and practised as the knowledge acquisition process proceeds. One conceptual proposal by Kathy Eckman is to organize the customer profile hierarchically along some pertinent dimensions so that customers differing on those dimensions will have different detail profiles. We will have to discover what these "pertinent dimensions" are from the subject matter experts.

7. *Where/who can we extract the knowledge from?*

Rationale:

The possible sources of knowledge include marketing units (owners/operators and managers market), product managers, sales organization (CCEs and management), RMIE database unit...

Remarks:

In the initial phase, we will concentrate on marketing units, product managers and CCEs. Different subject matter experts have to be identified and the extracted knowledge can be cross validated. Points of contact from marketing unit and product management will have to be identified.

4.6 Recommendations and Technology Options

For the sake of simplicity, recommendations will be divided into several areas. Each area is supported by the underlying reasons and goals.

Functional Requirements

One of the major organization/management-originated functional requirements is the capability of generating product recommendations from customer information and needs with benefits and counters. An effective and experienced CCE (in consultative selling) is competent in giving product recommendations from the analysis of customer needs. This proposed feature is a handy reference for the CCEs and can suggest alternatives to sell products. However, it should not form the only functionality of TIPS. The value this feature can add to TIPS is high but is not the critical element of success in consultative selling. Instead, traffic, usage and toll information of the phone lines of customers is among the most convincing substantiations of recommendations. Although this will be out of scope for TIPS at this point in time, it is definitely one of the most important functions of a sales helper system.

Because of the scarcity of information and the possible lack of quality knowledge for the knowledge base of the engine in the initial phase, the recommendations the system generates may not be accurate. In order to guarantee success in the project, other useful functionalities will have to be built into the system. The performance of the engine will improve over time as more data are transformed into knowledge of the domain.

To add more value to the project in the future, some of the following functions can be incorporated into the system:

- automation of SIC - provide a robust entry system using English instead of the code number;
- advertisements, blue-light specials and office blitz information;
- expansion of SIC to include brief descriptions of business procedures of customers;
- new product announcement;
- personal sales and revenue profile of CCEs;
- inter-departmental directory;
- regional current affairs bulletin board;
- closing date of directories;
- central office and equal access conversion dates;
- local calling areas;
- tax tables for each area

In the subsequent phases, most of the above functions can be incorporated into SSNQ without impacting on TIPS.

Technology Options

Because of the lack of empirical and usable information and the relative low ranking for statistical information across different functional groups or industries (ranked 15 out of 18 functions), the expert system technology will be more appropriate in the initial phase of the project. Marketing research analysts from strategic marketing of US WEST Inc. to owner/operator and manager market of SBS expressed the lack of discernible quantitative relationship that we can make use of to generate recommendations. They also stipulated that the data available are limited and difficult to make use of.

The owner/operator marketing personnel stopped using functional groups in their market analysis. From the focus group meetings of the CCEs, it is found that market segmentation (owner/operator vs. manager market), functional groups and SIC information are not used by the CCEs to give recommendations. Statistical information across different functional groups or industries does not receive a high ranking either. Instead, the CCEs give recommendations based on the unique situations and business needs of the customers (and the compensation scheme). One possible explanation offered by the client about the infrequent use of market segmentation and functional groups information by the CCEs is that this type of information is already embedded implicitly in the answers of customers to the questions of CCEs. The CCEs are using the information without any awareness.

It will be useful to know the typical telecommunication needs of the various types of business, but not the quantitative information that shows the distribution of products. Most of the customers of SBS have 3 lines or under and there is so much variation within a category of business that it makes little sense to produce distribution data directly to CCEs. According to Laird Brown, the marketing unit of the owner/operator does not make use of the quantitative information of functional groups in their market analysis either.

In order to train a neural network (using back propagation algorithm, which normally produces good results for problems involving predicting trends with quantifiable data) for the job involved, a relatively clean and complete data set should be used. Otherwise, the predictions it generates will not be accurate. Since we do not have data on the business profile/needs of the customers (we need quantifiable measurements of specific business profile/needs information associated with specific customers, not general statements about the needs), it will not be appropriate to use neural network as the underlying technology. The introduction of new products may also pose problems for the net because there will not be any examples to learn from. Another concern on the use of neural net is the lack of

explanations when the recommendations are generated. Neural network looks for patterns among the data but does not provide the reasons why there are such patterns.

The lack of quantifiable and measurable data of business profile/needs information associated with customers also means that we cannot use statistical analysis to establish any quantitative relationship between product utilization and customer profile.

From the above analysis, we can rule out neural network or statistical modelling as viable technologies as they do not fulfill our user requirements. However, we will keep these options open as tools to analyze the data we collect as they become available. When enough data are gathered and analysis of such data yields meaningful results, we can integrate quantitative information into the expert system framework.

Design and implementation strategies

The design of the engine should be flexible enough to integrate disparate sources of information, subject matter expert knowledge, empirical and historical knowledge on the customers profile and products used as well as traffic usage and other pertinent source of information. Modules like statistical analysis or neural network can be hooked or embedded into the expert system engine when quality data become available to serve as input to the decision process.

Although the "sales assistant sub-module" is important to provide more value to the system at this point in time, it should not form the major function of the system. Most of the functionalities addressed by the sales assistant sub-module belong to the domain of SSNQ and should be incorporated into SSNQ in the long run.

It will be extremely fruitful to include some top-performing consultative CCEs in the knowledge acquisition phase and incorporate their input into the engine, since they are involved in the day-to-day sales consultation and will be using the system. Their experience and capability will be customer-oriented. They excel in the analysis of the needs of customers and formulation of product recommendations. To save time and effort, two structured 1-day (or 2-day, if possible) workshops (of four to six participating CCEs) can be set up to elicit their knowledge at the beginning and towards the end of the knowledge acquisition phase. Four to six CCEs will be needed for more exhaustive and complete knowledge acquisition and cross validation. The subject matter experts of SSNQ and TIPS (user representatives Mike Henry and Cathe Valdez) will participate in the planning, as well as the knowledge acquisition phase of the process.

Although it may not be in the scope of the project in the first phase, subject matter experts of the marketing unit from the managers' market can also be actively involved in the knowledge acquisition process. The manager market constitutes less than 30% of the line accounts, but it generates more than 60% of the total revenue³ (9). CCEs in the business offices will have to deal with those accounts from time to time.

The involvement of different subject matter experts from different organization units is important because it gives a fair representation of the users of the system and provides cross validation of the knowledge used in building the knowledge base of the system. Since the required knowledge is distributed among subject matter experts in various organizations, the validity and accuracy of the knowledge base will increase as the "breadth" (number of subject matter experts) and "depth" (degree of participation of subject matter experts) of the project.

Market Analysis Tools

Bryan Meincke indicated that he used spreadsheets (Lotus 1-2-3) to analyze the data he got from RMJE. It would be much more productive for him to have a statistical analysis package on the PC platform for such purposes. Among the best packages found in the market are SPSS for Windows and Systat for Windows (as rated by PC Magazine) (10). However, Systat may not be suitable for large corporate data sets (like large survey or marketing data set) because of the limitations in data-management. Another choice may be SAS version 6.08 for Windows. It will be released soon but has steep pricing and licensing fee. To the very least, a spreadsheet capable of doing more statistical analysis (Microsoft Excel) should be very useful. Another area of consideration is the use of neural network programs to conduct exploratory analysis when data are available through TIPS. One package that can be used is the NeuralWorks Professional by NeuralWare. The prices and the learning curve are steep but it is a very good market research tool to have.

4.7 Living Laboratory Trial Issues

A "living laboratory" trial is scheduled in Denver for TIPS in December, 1993 after the initial trial of the customer manager module of SSNQ in November, 1993. The trial may pose problems if the data services for TIPS provided by SSNQ are not available at that time. Because of the

³ Owner/Operator Market consists of small businesses which have an urgent need to manage time to balance a variety of business activities while satisfying the personal needs of the owner. Manager Market consists of complex small businesses which have an urgent need to manage volume in order to expand and grow the business and/or increase efficiency in reduce cost and overhead.

dependencies on SSNQ, TIPS will not be able to meet its functional requirements if the existing services of customers and product information are not provided by SSNQ.

If it is found that SSNQ cannot provide the essential services at that time, we suggest that the trial date can be pushed out to 1994 when those services become available.

4.8 Design Constraints

The design of TIPS will be constrained by SSNQ. The information used by TIPS depends on the data services provided by SSNQ. The design of the user interface component will be dependent upon the view/controller/model architecture that SSNQ uses. The amount of screen estate available is also a problem. Since it will be very difficult for SSNQ to integrate TIPS into the product manager or account manager, it is very likely that TIPS will use separate window(s) for its functions.

One strategy we can have is to define what kind of data/information we need to display and gather and present our needs to the user interface developers of IT. In turn, they will consider the information and data requirements of TIPS and propose a scheme of implementation most suitable to their software architecture. They can prepare a prototype of their scheme and we can take it back to the users for suggestions or usability tests. Feedback from users can be taken back to the developers to negotiate for a better user interface. The process of having the developers of IT to propose a plausible design layout for TIPS will be more beneficial and constructive to the project given the stage that SSNQ is in.

4.9 Resource Estimation

The resource estimate below is done from a very preliminary basis. It represents only the personal estimate of the author.

It is difficult to estimate resource requirements for a project dealing with a vaguely defined domain. The most challenging part is the underlying engine and knowledge base construction of TIPS. At least two knowledge engineers from CGI will be required to capture and validate knowledge from the subject matter experts. Another team of engineers from CGI will be working on the engine and the knowledge base organization. Since the number of panels for TIPS will be relatively small and the design of panels will be constrained by SSNQ, the role of a human factors engineer will be limited to help to negotiate between the users and the developers to arrive at a design agreeable to both. Another part that a human factors engineer can play a role in this project is to act as a watchdog for the knowledge acquisition process by participating in the knowledge validation sessions and conducting usability testing of the prototype system. The human

factors engineer can also participate in the knowledge acquisition process by conducting studies on the knowledge obtained. It may also be appropriate for the Knowledge Systems Group (Science and Technology) or Quality Control Group to be involved.

At least two engineers from IT will be involved in the project. One is to develop and implement the user interface portion of TIPS; the other is to provide the necessary data services required by TIPS.

Resources required from the client organization will be the subject matter experts from CCEs, marketing and product management. In each of the organizations, at least one key contact should be established and participate as a team member. Training and documentation developers will also be needed. Contact points with other organizations that TIPS may have an impact on (or have an impact on TIPS) need to be identified.

Another important resource requirement is the personnel from SBS to maintain the knowledge base and data base of the system. Without constant updates, the system will not be able to provide timely and consistent sales information and tips for CCEs. The identification and assignment of staff to this function is vital to the success of the project in the long run.

4.10 General Conclusions and Comments

Although TIPS in the initial phase may have inadequate knowledge of customers to generate accurate recommendations, it can potentially become one of the most valuable assets an organization possess - a knowledge base of its customers. The business value it generates will be enormous as more and more data and knowledge of our customers can be gathered. With the coordination of market analysis and research, the benefits reaped will be significant.

Initially, there will be trial-and-error efforts expended to explore and organize the knowledge distributed among the subject matter experts. These efforts should be well spent because it creates a synergistic effect on the process of knowledge acquisition and knowledge organization. Not only that we can understand more about the problem domain to implement an effective system, the subject matter experts can become more conscious of their organization of knowledge and may eventually be able to present knowledge in a sharable and easily understood form.

In the short run (within a time frame of a year or half a year), the risk of this project is high because of the uncertainties involved in SSNQ and the detail of knowledge of the subject matter experts has. Once the feedback loop between marketing organization and CCEs is established, it can create a resonant effect on the knowledge gained from TIPS.

Last but not least is the cost-benefit consideration of the project. Again, since the problem domain is complex and relatively vague, it is difficult to put it into perspective. From the users' standpoint, there is definitely a lot of value to this project (particularly on the customer information and profile). However, the actual cost of delivering the functional requirements for the project should be weighed carefully against the value added. Admittedly, this project carries a high risk given its dependency on the knowledge of subject matter experts and the support from SSNQ. However, if we can make good use of the information collected and improve the knowledge base continuously, TIPS will form one of the most useful tools not only for the CCEs, but for marketing and product management as well.

5. Issues

The issues that may impact the quality and schedule of the project are divided into major areas and enumerated as follows. Brief suggestions by the parties involved as to how to address those issues are included if possible:

Organization/Management-originated vs. user-originated requirements

Issue	Impact	Suggestion from Users
WRAP Strategies	CCEs may not be able to serve unique customer needs if recommendation is based on compensation objectives based on WRAP Strategies	Provide better and effective implementations of WRAP Strategies; Evaluate usefulness of WRAP Strategies
Market Segment and Functional Group	CCEs make little use of the information	Explore use of other pertinent information; Evaluate usefulness of information
Office Objective	CCEs think that office objective is incompatible with consultative sales	Reformulate objectives to support consultative sales

Table 1:

User-originated requirements vs. Project Scope

Issue	Impact	Suggestion
Usage and Traffic patterns of customers' phone lines	CCEs use this information to substantiate their recommendations	Provide such information in the next phase of the project

Table 2:

SSNQ and TIPS Issues

Issue	Impact	Suggestion
SSNQ may not be able to provide data services and user interface implementation support	TIPS schedule delay	possible contingency plan
TIPS may not be able to integrate with SSNQ as planned	Inconsistencies in the design of TIPS and SSNQ may further the integration problem	
User interface and data services of TIPS constrained by design of SSNQ	Unnatural and confusing screen design of TIPS to accommodate design of SSNQ	

Table 3:

Addendum to Final Report

The release of the final report has been postponed to June 30, 1993 upon the request of Carnegie Group Inc. to include their feedback on the pre-test draft of this document. No other changes have been made to the contents of the final report.

The findings of this report corroborate with the TIPS Focus Group Project Report, prepared independently by Norma Emond, who served as a facilitator for the groups and the DARWIN Phase 2 Report prepared by Mitch Glasser. In a nutshell, the information needs of the CCEs we found in the focus groups also reflect the needs and expectations of our customers. This coincidence is by no means accidental. The CCEs work in the forefront of customer service and relations. In order to promote customer loyalty and improve customer service, we need to involve them as a major source of input for the system we plan to build. Their knowledge of the customers' needs and the market cannot be overlooked.

Meanwhile, US WEST IT is reviewing what they will deliver for TIPS. There are still a lot of open issues of TIPS and SSNQ. We will expect to work those issues out in the next 2 or 3 months.

6. References

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Appendices

See attached documents:

1. Management Interview Questions
2. Marketing Interview Questions
3. Rating for Focus Group
4. Reply and clarification from Readers
5. Reply from Kathleen Eckman (CGI) dated 6/29/93

Management Interview Questions

1. Based on the high level vision of TIPS that you just saw, we need to know what are the business needs or problems, specific to your organization, that we are trying to solve?
2. Who would be the targeted user- top-end sellers, average sellers, or low-end sellers? Why?
3. A. On a scale of 1-5 rank the following in order of importance:

NORMAL BUSINESS CONDITIONS	UNDER ALERT STATUS	Functions
_____	_____	CUSTOMER SERVICE
_____	_____	SERVICE QUALITY
_____	_____	SPEED OF ANSWER
_____	_____	# OF OUTGOING CALLS MADE
_____	_____	# OF INCOMING CALLS RECEIVED
_____	_____	SALES

B. During our business office visits we noticed the above behavior drivers. What type of behavior should TIPS drive?

4. We know what our customers want based on the "Voice of the Customer" survey that was put together by Shaun Mee and we know competition is at our heels, along with that we have our "Wrap Strategies" - Our objective is to enable our sales consultant to focus on our customers in a manner where they won't go to our competition. Can you give us your thought on what kind of behavior that should be?
5. Do you have any idea what features or benefits TIPS could provide that would add value to motivate a top performer?
6. We have visited the business office and checked with Ameritech who developed a system similar to TIPS. Do you have an idea on how we can get the Sales Consultant to buy into a system like TIPS?
7. In your view, are there or will there be a range for the length of a consultative sales call?

8. We have product strategies, marketing strategies and competition strategies. What strategy would you like met from a system like TIPS?
9. A. What kind of marketing intelligence information would help in the sales consultant's job?
B. What kind would your organization want?
10. Should a system like TIPS key in on our Flagship Products or all 261 products US West offers?
11. A. How do you see the sales consultant's job changing as competition for Small Business customers heats up?
B. How often do you think we will have to adapt our tactics?

Marketing Interview Questions

3-1-93

revised 4-16-93

We plan to develop Customer Analysis and Sales Helper as a software tool to be used with the Sales and Service Negotiation and Quality System. This system will enable the sales representatives to provide recommendations on products based on customer profile and current market conditions. In the process of sales consultation, the sales representative will also gather valuable information from the customer for the marketing unit for further analysis and research.

We are at the stage of developing user requirement analysis and would like to choose a technology platform best suited to the requirement. We need your honest and valued input to this project. With your assistance and advice, we hope that the new tools and technology introduced for this project can be reused and will enhance the analysis of market and customer data.

1. What kind of approaches / technology have been used by the marketing organization to analyze the relationship of sales of products, customer / account profiles and current market conditions? Describe how you analyze your data and generate your market knowledge?
 - a) What are the sources of your data? How accurate are your data?
(4-16-93)
2. Do you feel that quantitative relationships between customer characteristics, market conditions and the telecommunication services bought exist? Can you define such relationships?
3. How much data do you have in the above-mentioned arena? Can you give us some details on the nature of the data, like what kind of variables you are looking at or collecting?
4. What kind of data analysis have you carried out on the data? Can you identify any correlations between your variables? What kind of further analysis do you plan to do with your data?
5. Can you articulate or formulate your knowledge about the market and the customer profiles (including the products they buy) in terms of decision rules? How many decision rules do you have to formulate in order to have a non-trivial system of recommending products to customers of different profiles? What degree of accuracy will you anticipate such rules to be? How often will you expect to change the rules in response to different market conditions? How will you organize and communicate your knowledge to the sales representatives?

6. If you cannot articulate your knowledge about the market and customer profiles explicitly in terms of rules, how do you anticipate organizing and communicating your knowledge to the sales representatives?
7. Is knowledge of relative or absolute frequency of specific purchased products and services among a customer category useful to sales representatives? What is the granularity of "customer category" will you suggest? (4-16-93)
8. Most of the sales representatives in our focus group meetings express that each customer is unique and their needs depend on their business situations. they do not make use of functional group or market segment very much to make recommendations. they think that functional group and market segment information is not useful in consultative selling. what do you think? (4-16-93)
9. What kind of customer information do you like to gather from the sales representatives? Given the nature of their job, we have to keep the amount to a minimum with impairing their job flow. (4-16-93)
10. What kind of market intelligence information do you like to have? How are you going to gather this?

Rating for Focus Group

Ideal Sales Helper Functions

Functions	High	Med	Low	Score	Rank
sales tips from profile	12	6	2	2.50	5
past toll (12 months)	11	7	3	2.38	8
product statistics	7	10	3	2.20	15
benefits & counters	11	6	3	2.40	7
advertisement, special	13	6	1	2.60	3
account profile	15	2	2	2.68	2
how products work	12	3	6	2.29	10
daily feedback	11	3	7	2.19	16
descriptions of business	8	5	8	2.00	18
USOC (English) catalog	7	5	3	2.27	11
weekly summary of bulletin	8	6	1	2.47	6
intra-department directory	8	3	4	2.27	11
future product availability	7	7	1	2.40	17
directory close dates	13	1	1	2.80	1
current affairs in region	6	7	2	2.27	11
feedback channel	6	5	4	2.13	17
conversion dates	8	4	3	2.33	9
local calling areas	11	2	2	2.60	3

Response from Kathleen Eckman 6/29/93 1:51pm

Introduction

The document entitled *Target Information and Profile System Recommendation and Evaluation of Future Direction* touches on several areas that should be explored in more depth before conclusions are finalized. Included in this analysis are discussions of culture changes, knowledge based systems, the focus groups, and the Home Office Trial.

Paradigm Shift

The Customer Contact Employee (CCE), formerly the service representative, is facing a fundamental paradigm shift from being an order taker to being a consultant. This shift is taking place because U S WEST is facing the loss of significant market share due to the entrance of competition into the telecommunications arena. To be competitive, CCEs can no longer be only order takers. They must be able to effectively consult to provide telecommunications expertise to U S WEST customers.

The referenced document addresses strictly the present method of operation, implying the today's culture and attitudes will not change. The Target Information and Profile System (TIPS), must work in this environment, but more importantly, in the future method of operation. TIPS provides one mechanism to help the CCE make the paradigm shift, while also providing benefit to the customer and to the corporation. In order to be successful, TIPS must be built with an eye to the future.

In order to serve the CCE, the corporation and the customer, more analysis must be done. The focus groups plus TIPS' client leads serve as the CCE representation and are discussed in depth in the document. More in-depth discussions about management (i.e. the market unit) and customer issues should be included. These issues are concerned with competition, loss of market share, cultural changes in the business office of the future, and customer needs. All concerns, the CCEs', the corporation's, and the customers', must be considered when developing TIPS.

Knowledge-Based Systems

"Expert Systems" are so termed because they are developed by extracting knowledge from experts. This knowledge is typically not "easily elicitable", not robust, not quantitative. Instead, such knowledge is qualitative, generated by years of on-the-job experience, and is typically heuristic in nature ("rules of thumb"). Because this knowledge has been abstracted and internalized, experts are generally not even aware of how they do it! Direct questions to get at their thought processes are often not very

effective; the knowledge-engineering process to extract this knowledge from experts is often painful. Seldom is the knowledge quantified. Despite these characteristics, successful expert systems have been developed. It is part of the signature of an expert system that it provides reasonable output on even limited inputs and information.

The knowledge engineering required for the TIPS project is not trivial. The situation is further complicated by the possibility that even the experts may not be doing the job as expertly as desired. However, if the knowledge engineering efforts are supported, the initial phase of TIPS will be able to respond with good recommendations. It is important to collect information from the selling and marketing angles in order to take into account both current and anticipated practices. Furthermore, the ability to analyze the profile data collected by TIPS and further refine the knowledge base will ensure the long-term success of the system.

Without good knowledge engineering, the recommendations made by TIPS will probably not be very good. Even with good knowledge engineering, the problem is complex enough that some cooperation with the human users may be needed to supply judgements not easily derived by machines. The system will continuously improve through the incorporation of the suggestions of superior performance users into the knowledge base that is used by all CCEs.

Focus Groups

The focus groups conducted with CCEs were an important step in gathering user requirements. Care must be taken, however, in summarizing and interpreting the results. Some of the suggestions received need further analysis before inclusion in any tool. For example, how is toll information used? Does the CCE need toll information directly, or would an automated analysis of the information, resulting in product recommendations or sales support, be more applicable? Similarly, traffic study information was deemed useful, but the real benefit may be in automatically combining traffic study results with other profile items. Instead of supplying tax tables to the CCE, it may be better to incorporate that information so the CCE does not have to read a table. Before automating SIC entry, it may be advantageous to explore the usages of SIC, and the possible creation of more meaningful classifications. These are examples of good ideas generated in the focus groups that warrant further study.

The WRAP strategies were also discussed in the focus groups. It is important to distinguish between the CCEs' view of them and management's view of them. The CCEs view WRAP strategies as a compensation plan, while management views WRAP strategies as a strategy to maintain market share with competition entering the

environment. The WRAP strategies make sense at the strategic level to the CCEs, but the compensation for the strategies are responsible for some of the CCEs's discontent.

Home Office Trial

The Home Office Trial, a joint venture of H&PS and SBS, was conducted in Salt Lake City from October 12, 1992 through April 2, 1993. It was extended past the original three months for many reasons, including inadequate training of the consultants. The usability study done by AT was conducted very early in the trial, which impacted the results of that study. There were also changes to the CASH interface for the second three months of the trial, but no further usability tests were performed.

Of the eight participating service representatives, six of them had never had any consultative training. In addition, they had never used a windowing system or a mouse prior to the training that occurred just two work days before the start of the trial. There were inadequately trained in the home office market segment, mostly because the training was concentrated on product training. System training was limited to four hours for CASH, and four hours for COMPASS.

Inadequate systems training plus no consultative training greatly impacted the usability of CASH. It is possible that recommendations were not useful because the service representatives were unable to manage the mouse and windows to navigate the system to get the recommendations at the proper time. Many of the service representatives did compare their own recommendations to the system's recommendations after the completion of a call, using the system as a training aid.

Another impact on the usefulness of the recommendations is CASH's inadequate handling of multiple lines. Besides not knowing how many lines a customer had and what services were on each, CASH was originally designed to recommend most products on a single line (it was built for H&PS where this is most common.) As TIPS is developed for SBS, recommendations for multiple lines is critical.

There was positive feedback about CASH as well. Being able to save and recall a customer's profile was highly valuable to the service representatives. The H&PS representatives, although they had difficulty navigating through the profile items in the early stages of the trial, found that the display of the profile items taught them which questions were important to ask.

It is important to mention that the culture of the Home Office Trial was different from the BSC or RMC culture. For example, people called into this channel because of a free booklet and consultation offer. The

representatives were not compensated on any sales objectives, but on a satisfaction survey conducted on customers who used the service. These cultural differences should be taken into account when comparing these representatives with CCEs in the BSC. Although the usability study was performed far too early to measure the effect, the Home Office Trial demonstrated the use of the tool to facilitate a shift in culture to true consultative selling practice. The verbal responses of the CCEs closely matched the knowledge base in the final months of the trial.

Conclusion

The document entitled Target Information and Profile System Recommendation and Evaluation of Future Direction provides an in-depth analysis of today's business office and current CCE practices. This analysis summarizes valuable information about the business offices, but needs to address larger and more complex issues when analyzing the future of TIPS.

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Organization	Individual
USWEST SBS	Ed Barney
USWEST SBS	Cathe Valdez
USWEST SBS	Mike Henry
USWEST SBS	Diane Boileau
USWEST IT	Judy Warner
USWEST IT	Gary Craig
USWEST IT	Jim Bestem
USWEST AT	Susan Wagner
USWEST AT	Gail Fontenelle
USWEST AT	David Sward
USWEST AT	Chris Plott
CGI	Kathy Eckman, Alan Lawson

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